THE ABSOLUTELY WEAKLY NORMAL STRUCTURE SHEAF IS A SHEAF FOR THE PH TOPOLOGY

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file flat.tex; no statement or proof step was rewritten, reordered or removed.

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Editorial note. The counted statement and proof contain no cross reference, no citation and no included file, so the delivered body is self contained.

Editorial note. Canonical location of the counted proposition: Stacks tag 0EVT (sources/stacks-flat/pinned-original-source.tex line 11991).

Allowed standard prerequisites (editorial).

- The characterisation of ph sheaves by two conditions. Topologies, Lemma labelled topologies-lemma-characterize-sheaf, cited at proof lines 12001-12003. Reduces the sheaf property to the Zariski and the equalizer conditions.
- Commutation of the absolute weak normalization with open coverings. Morphisms, Lemma labelled morphisms-lemma-seminormalization, cited at proof lines 12005-12008. Settles condition (1).
- Scheme theoretic images of proper morphisms and finiteness of proper morphisms with affine target. Morphisms, Lemmas labelled morphisms-lemma-scheme-theoretic-image-is-proper and morphisms-lemma-finite-proper, cited at proof lines 12020-12027. Produces the closed subscheme Z of A^1_X proper and finite over X.
- Universal injectivity and universal homeomorphisms. Morphisms, Lemmas labelled morphisms-lemma-universally-injective and morphisms-lemma-universal-homeomorphism, cited at proof lines 12060-12068. Converts the pointwise injectivity proved by the field argument into the universal homeomorphism applied at the end.

Source reference. Original definition 0DBL: sources/stacks-chow/tags.txt, source lines 16701–16701. Cross-chapter standard prerequisite located in the project tag list staged in the local snapshot; invoked at proof line None of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

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Source reference. Original lemma 0DBQ: sources/stacks-chow/tags.txt, source lines 16705–16705. Cross-chapter standard prerequisite located in the project tag list staged in the local snapshot; invoked at proof line 12003 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original lemma 0EUS: sources/stacks-chow/tags.txt, source lines 18562–18562. Cross-chapter standard prerequisite located in the project tag list staged in the local snapshot; invoked at proof line 12005 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original lemma 0AH6: sources/stacks-chow/tags.txt, source lines 13219–13219. Cross-chapter standard prerequisite located in the project tag list staged in the local snapshot; invoked at proof line 12023 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original lemma 01WN: sources/stacks-chow/tags.txt, source lines 2488–2488. Cross-chapter standard prerequisite located in the project tag list staged in the local snapshot; invoked at proof line 12025 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original lemma 01S4: sources/stacks-chow/tags.txt, source lines 2325–2325. Cross-chapter standard prerequisite located in the project tag list staged in the local snapshot; invoked at proof line 12061 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original lemma 04DF: sources/stacks-chow/tags.txt, source lines 5725–5725. Cross-chapter standard prerequisite located in the project tag list staged in the local snapshot; invoked at proof line 12063 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Lemma 0.1. *Let $(Sch/S)_{ph}$ be a site as in Topologies, Definition 0DBL. The rule*

$$X \longmapsto \Gamma(X^{awn}, \mathcal{O}_{X^{awn}})$$

is a sheaf on $(Sch/S)_{ph}$.

Proof. To prove $\mathcal{F}$ is a sheaf, let's check conditions (1) and (2) of Topologies, Lemma 0DBQ. Condition (1) holds because formation of X^{awn} commutes with open coverings, see Morphisms, Lemma 0EUS and its proof.

Let $\pi : Y \to X$ be a surjective proper morphism. We have to show that the equalizer of the two maps

$$\Gamma(Y^{awn}, \mathcal{O}_{Y^{awn}}) \to \Gamma((Y \times_X Y)^{awn}, \mathcal{O}_{(Y \times_X Y)^{awn}})$$

is equal to $\Gamma(X^{awn}, \mathcal{O}_{X^{awn}})$. Let f be an element of this equalizer. Then we consider the morphism

$$f : Y^{awn} \longrightarrow \mathbf{A}^1_X$$

Since $Y^{awn} \to X$ is universally closed, the scheme theoretic image Z of f is a closed subscheme of $\mathbf{A}^1_X$ proper over X and $f : Y^{awn} \to Z$ is surjective. See Morphisms, Lemma 0AH6. Thus $Z \to X$ is finite (Morphisms, Lemma 01WN) and surjective.

Let k be a field and let $z_1, z_2 : \mathrm{Spec}(k) \to Z$ be two morphisms equalized by $Z \to X$. We claim that $z_1 = z_2$. It suffices to show the images $\lambda_i = z_i^* f \in k$ agree (as the

structure sheaf of Z is generated by f over the structure sheaf of X). To see this we choose a field extension K/k and morphisms $y_1, y_2 : \mathrm{Spec}(K) \to Y^{awn}$ such that $z_i \circ (\mathrm{Spec}(K) \to \mathrm{Spec}(k)) = f \circ y_i$. This is possible by the surjectivity of the map $Y^{awn} \to Z$. Choose an algebraically closed extension Ω/k of very large cardinality. For any k-algebra maps $\sigma_i : K \to \Omega$ we obtain

$$\mathrm{Spec}(\Omega) \xrightarrow{\sigma_1, \sigma_2} \mathrm{Spec}(K \otimes_k K) \xrightarrow{y_1, y_2} Y^{awn} \times_X Y^{awn}$$

Since the canonical morphism $(Y \times_X Y)^{awn} \to Y^{awn} \times_X Y^{awn}$ is a universal homeomorphism and since Ω is algebraically closed, we can lift the composition above uniquely to a morphism $\mathrm{Spec}(\Omega) \to (Y \times_X Y)^{awn}$. Since f is in the equalizer above, this proves that $\sigma_1(\lambda_1) = \sigma_2(\lambda_2)$. An easy lemma about field extensions shows that this implies $\lambda_1 = \lambda_2$; details omitted.

We conclude that $Z \to X$ is universally injective, i.e., $Z \to X$ is injective on points and induces purely inseparated residue field extensions (Morphisms, Lemma 01S4). All in all we conclude that $Z \to X$ is a universal homeomorphism, see Morphisms, Lemma 04DF.

Let $g : X^{awn} \to Z$ be the map obtained from the universal property of X^{awn}. Then $Y^{awn} \to X^{awn} \to Z$ and $f : Y^{awn} \to Z$ are two morphisms over X. By the universal property of $Y^{awn} \to Y$ the two corresponding morphisms $Y^{awn} \to Y \times_X Z$ over Y have to be equal. This implies that $g \circ \pi^{wan} = f$ as morphisms into $\mathbf{A}^1_X$ and we conclude that $g \in \Gamma(X^{awn}, \mathcal{O}_{X^{awn}})$ is the element we were looking for. $\square$